

```
# Experiment Configuration
Element Budget: 25
Gamma_C: 50.0
Max Episode Steps: 200
Entropy Coefficient: 0.01
Learning Rate: 0.0003
```

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# Adaptive Mesh Refinement RL Training Summary
Date: 2025-03-26 07:05
Total Steps: 50000
Episodes Completed: 1221
```

#### ## Termination Statistics

- Budget exceeded (pre-action): 59 (4.8%)
- Budget exceeded (post-adapt): 37 (3.0%)
- Error: Mesh quality issues: Element size ratio too large: 384.00: 24 (2.0%)
- Maximum episode steps reached: 6 (0.5%)
- Maximum consecutive no-actions reached: 1095 (89.7%)

#### ## Action Distribution

- Coarsen: 3543 (7.1%)
- No Change: 36102 (72.2%)
- Refine: 10355 (20.7%)

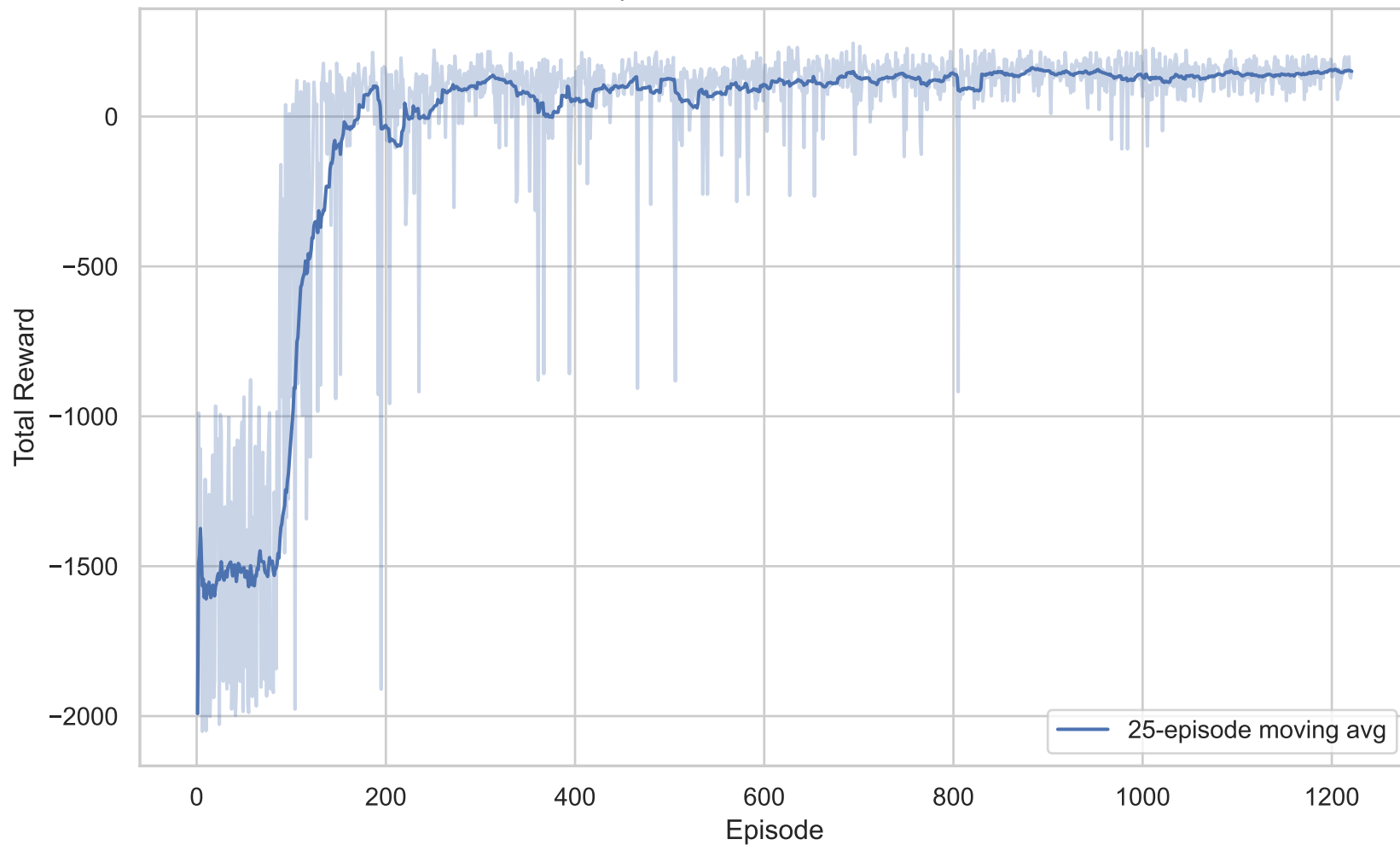
#### ## Reward Statistics

- Mean Reward: -31.74
- Min Reward: -2051.21
- Max Reward: 244.72
- Mean Reward (last 244 episodes): 138.10

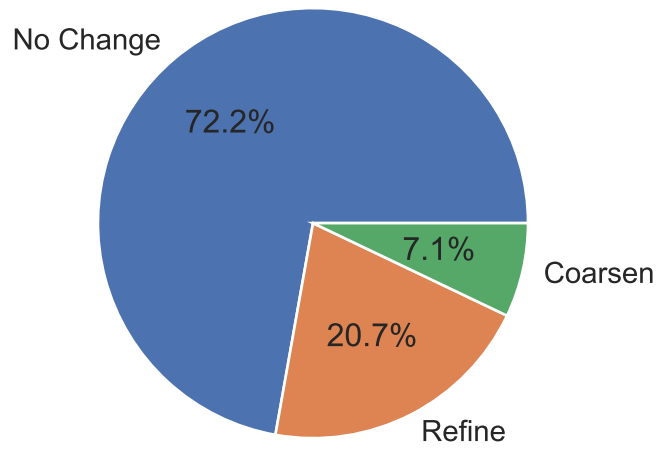
#### ## Resource Usage

- Mean Resource Usage: 0.46
- Max Resource Usage: 1.00
- Episodes ending with budget exceeded: 0 (0.0%)

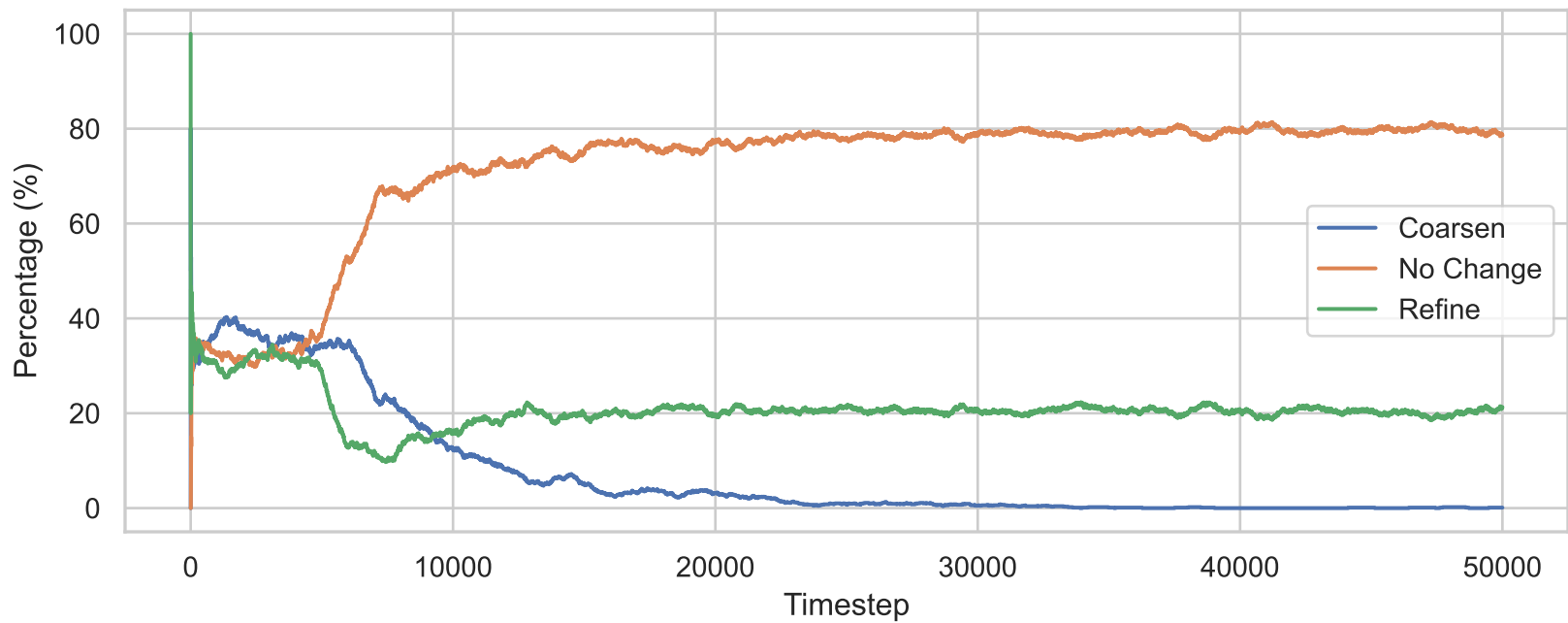
Episode Rewards Over Time

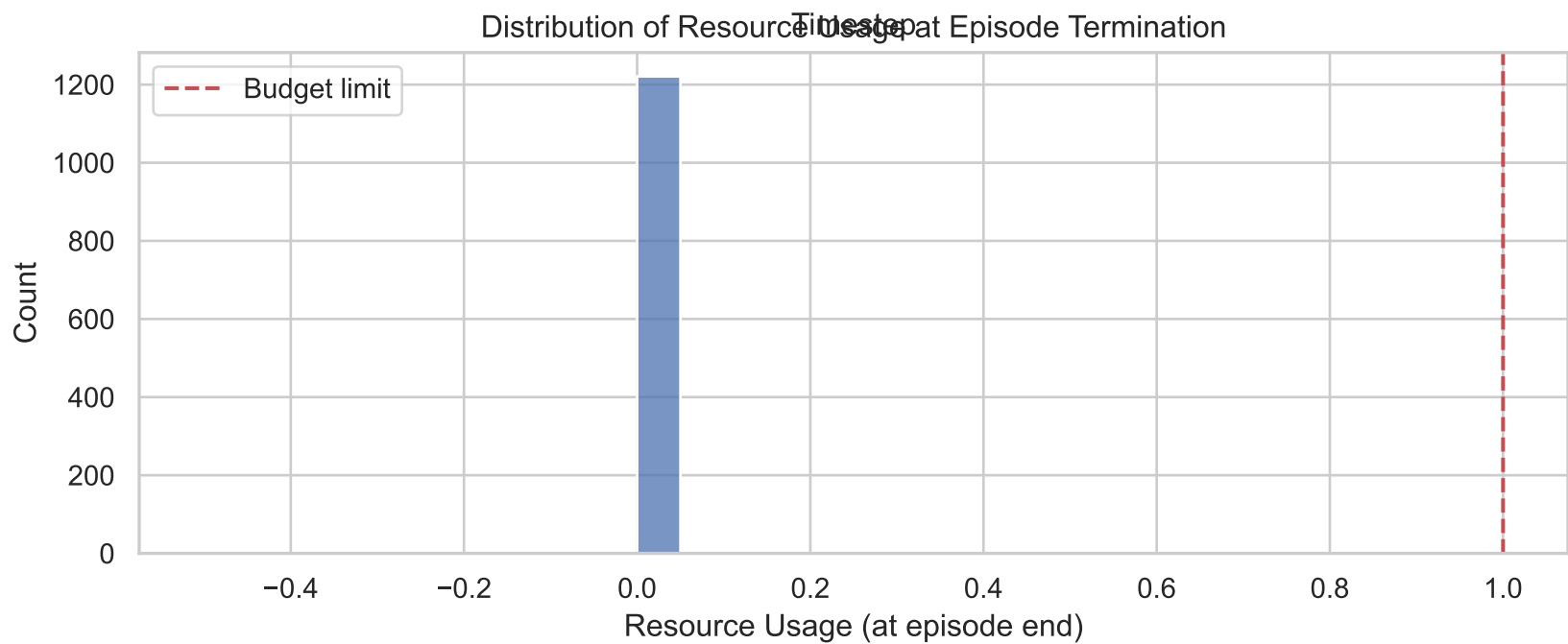
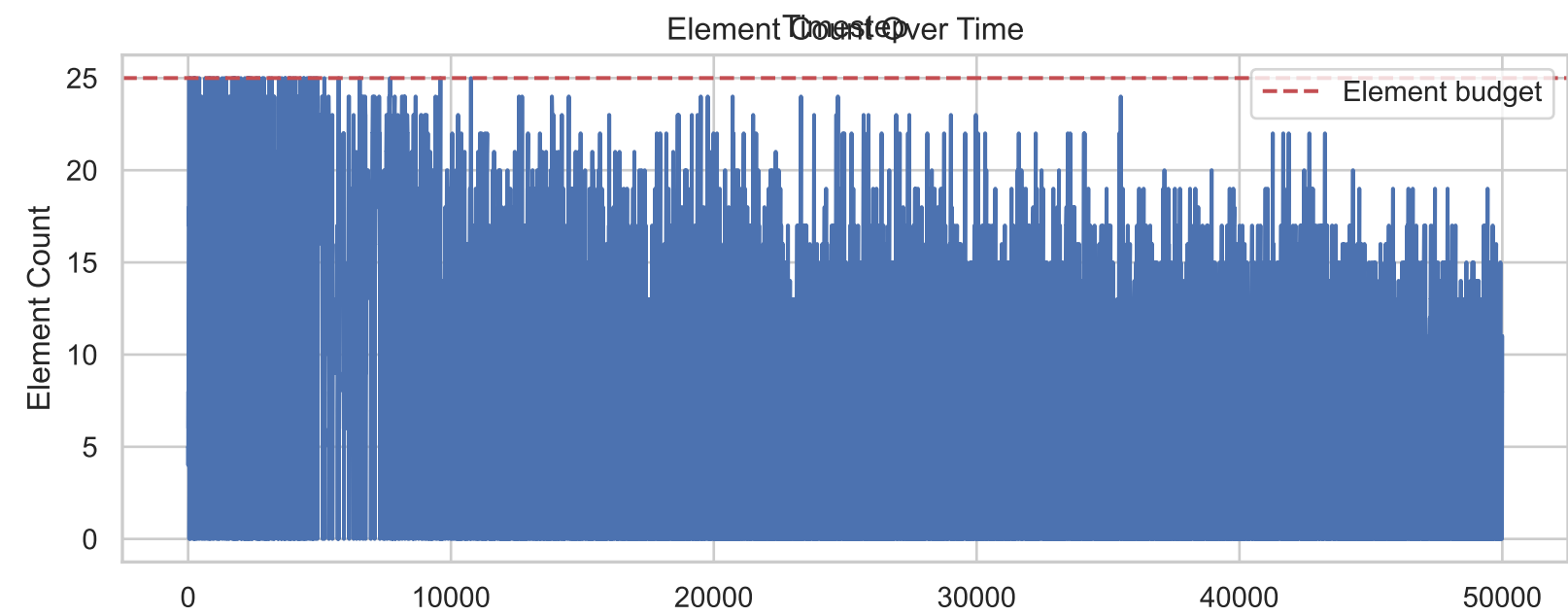
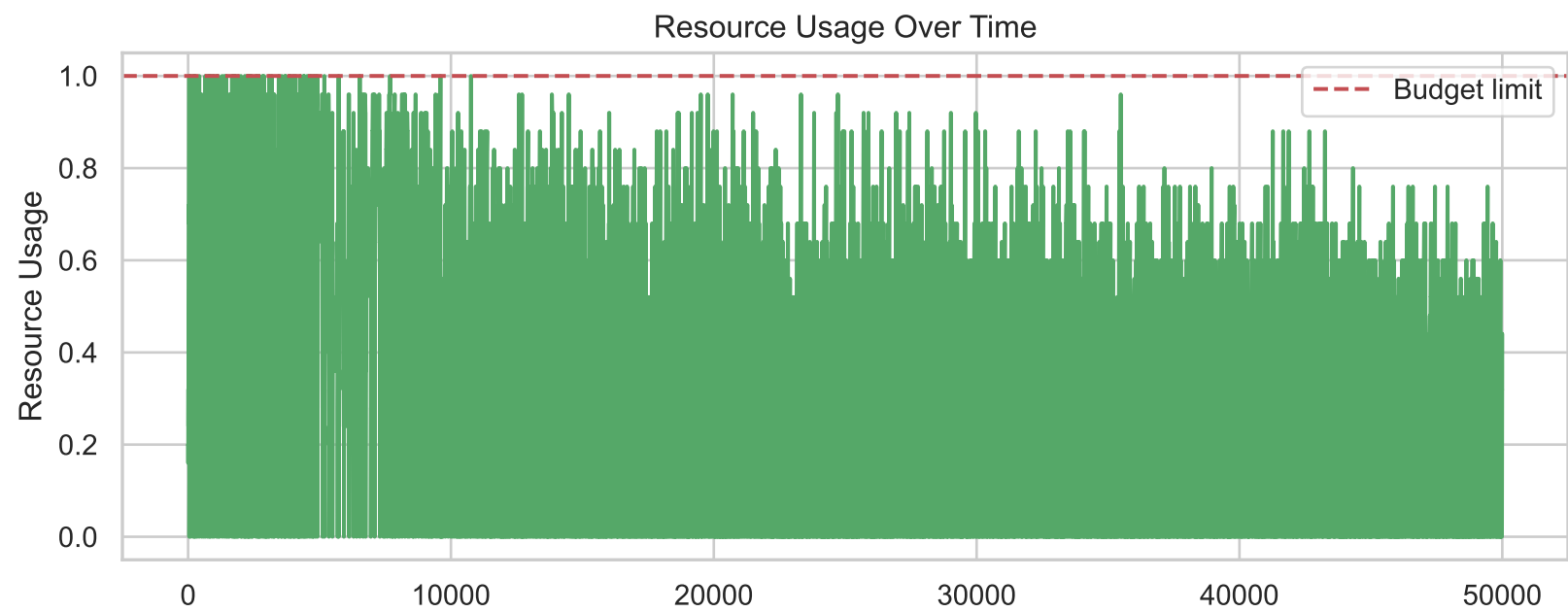


Overall Action Distribution

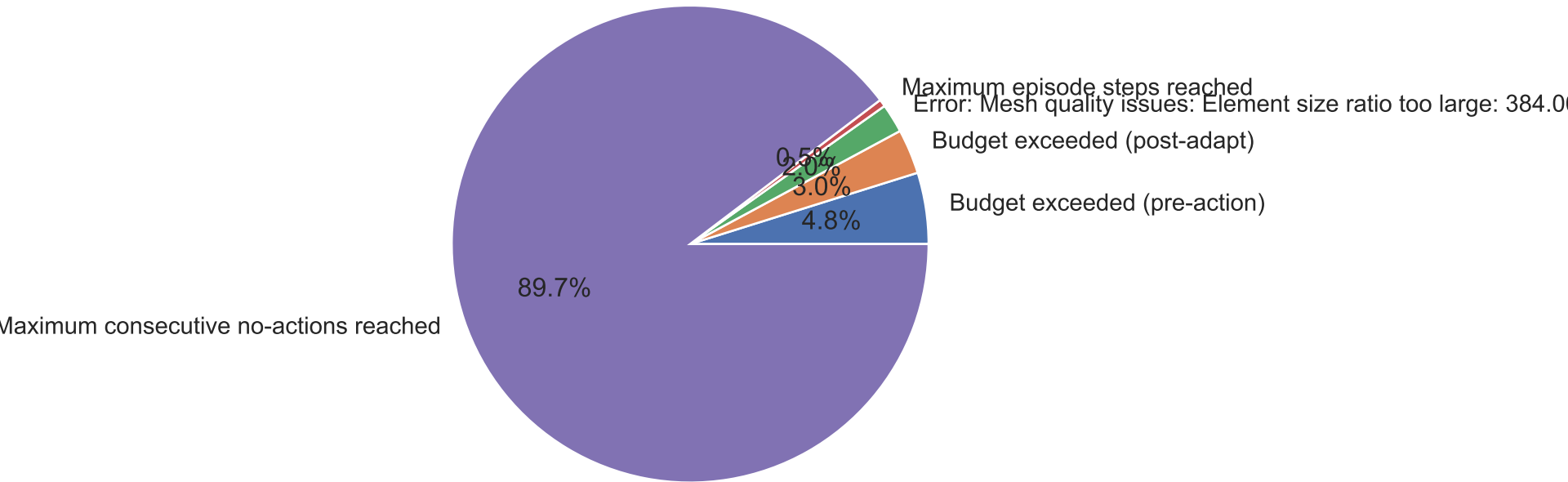


Action Distribution Evolution

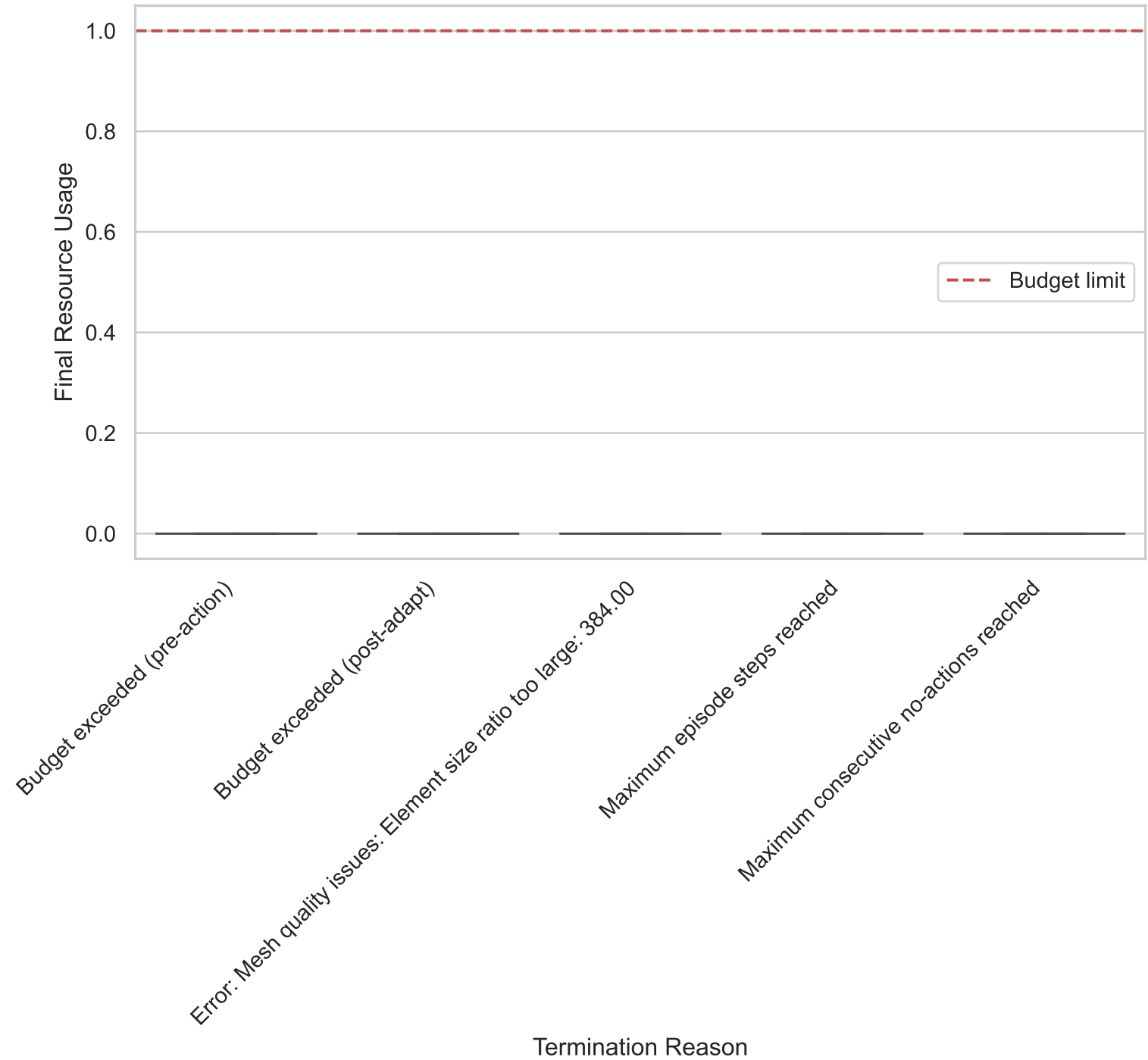




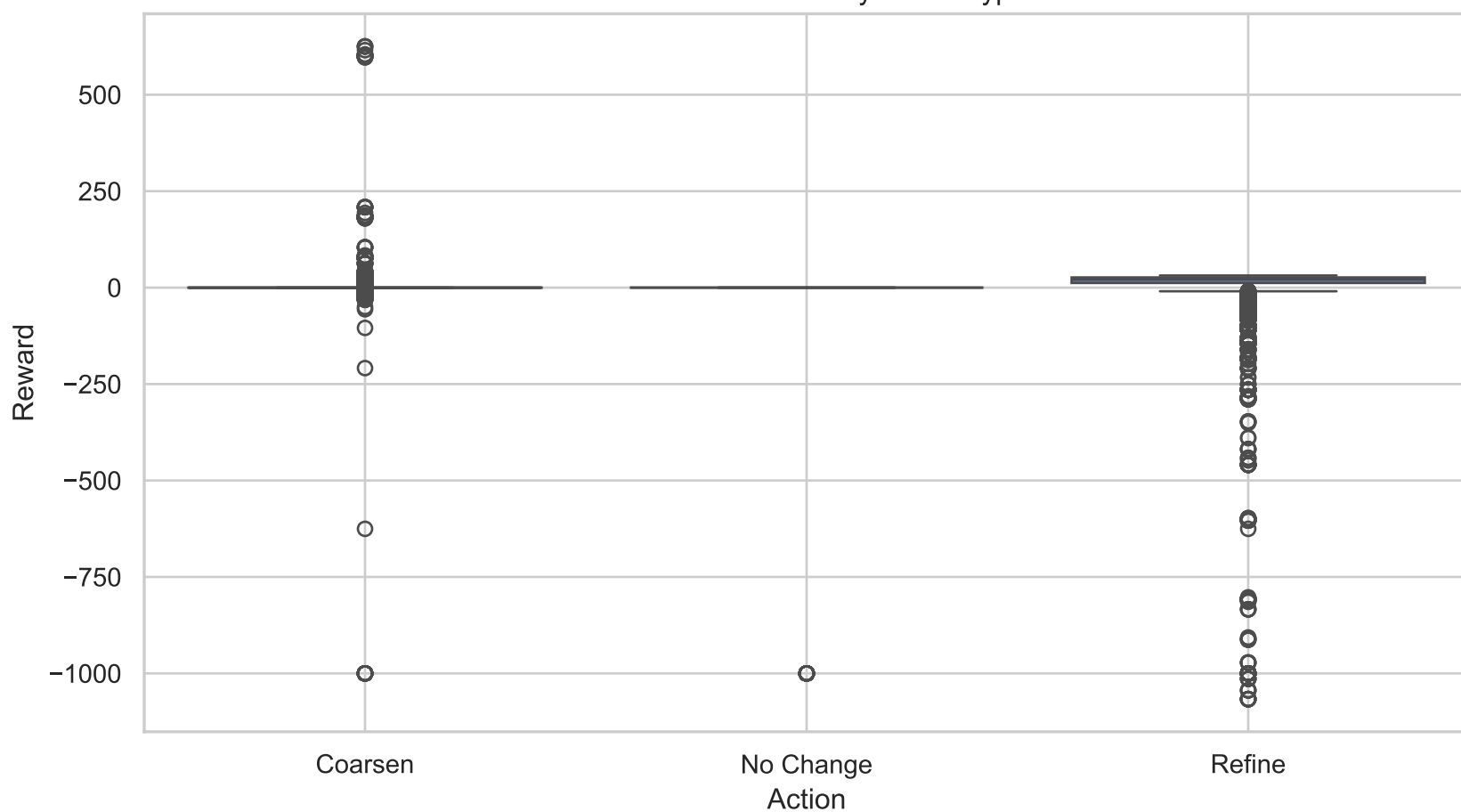
Episode Termination Reasons



Resource Usage by Termination Reason



### Reward Distribution by Action Type



### Episode Length vs Reward

